

DESIGN AND FABRICATION OF SETUP FOR HARVESTING WATER FROM ATMOSPHERIC AIR USING THE CONCEPT OF PELTIER EFFECT

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ABSTRACT

In numerous nations like India, accessing water resources for irrigation or other needs, particularly in arid regions, poses a significant challenge. Water scarcity is a prevalent issue worldwide, often stemming from insufficient rainfall. Conversely, in highly humid locales such as coastal areas, water can be procured through condensing atmospheric water vapour. This paper introduces a method for developing a water condensation system utilizing thermoelectric cooling technology. The system comprises cooling elements, a heat exchange unit, and an air circulation mechanism. Powered by a solar cell panel unit with a substantial output, the cooling elements are regulated through a control circuit. An Atmospheric Water Generator, a device capable of converting atmospheric moisture directly into potable water, operates on the principle of latent heat to transform water vapour molecules into water droplets. Although introduced earlier, it remains relatively uncommon in India and other regions. However, given the contemporary emphasis on renewable energy sources, its potential applications are considerable.

Keywords- Water condensation, Thermoelectric peltier, Dew condensation (latent heat), Solar energy.

I. INTRODUCTION

The Earth's atmosphere holds a significant amount of water in the form of vapour and moisture. Astonishingly, nearly 30% of this valuable resource is currently being wasted. However, this wasted water can be reclaimed through the implementation of innovative technology like the Atmospheric Water Generator [1]. This remarkable device has the capability to directly convert atmospheric moisture into clean, drinkable water. Operating on the principle of latent heat, the device efficiently transforms water vapour molecules into liquid droplets. Many regions across the globe, including areas in India, experience varying climates—from deserts to rainforests—and face challenges with water scarcity. Despite the abundance of atmospheric humidity in these areas, water resources remain limited. Over the past few years, several projects have explored methods of harnessing air condensation and water generation using technologies like Peltier devices, such as the utilization of Peltier plates powered by photovoltaic solar energy to harvest water for young trees [2]. This project aims to expand upon these advancements, furthering the applications of such devices in the foreseeable future. Traditionally, the temperature required to condense water is referred to as the dew point temperature. The objective here is to practically achieve this specific temperature through experimentation, utilizing electronic devices. The project involves the utilization of a thermoelectric Peltier (TEC) couple [3] to create the necessary environment for water condensation. Alternatively, a conventional compressor and evaporator system could also be employed for condensation by exchanging the latent heat of the coolant within the evaporator.

Peltier couple:

The Peltier thermoelectric device [4] operates on a fascinating principle involving two sides—a p-type and an n-type semiconductor. When a DC current is applied, heat is transferred from one side of the device to the other, resulting in one side becoming cooler while the opposite side heats up. This phenomenon, known as the Peltier effect and electron hole theory, forms the basis of Peltier coolers. These coolers consist of a Peltier element and a robust heat sink/fan assembly. Peltier elements come in various configurations, typically comprising numerous thermocouples arranged in a rectangular pattern and encapsulated between two thin ceramic plates. Such devices are capable of freezing a significant volume of water within a matter of minutes. Similarly to conventional cooling systems, a TEC (Thermoelectric Cooler) also comprises three essential components: the evaporator, compressor, and condenser. In a TEC, electrons absorb energy (heat) at the cold

junction as they transition from a lower energy state in the p-type semiconductor element to a higher energy state in the n-type semiconductor element. The power supply facilitates this electron movement through the system. Conversely, at the hot junction, energy is released to a heat sink as electrons move from the higher energy level element (n-type) to the lower energy level element (p-type). This intricate process enables efficient heat transfer within the device, making it a versatile tool for various cooling applications.

II. WORKING FEATURES

In practical applications, Thermoelectric (TE) couples are often combined into modules, electrically connected in series and thermally connected in parallel to maximize their output potential. However, it becomes impractical to utilize devices with lower work done to power ratios. The market offers a variety of TE modules differing in sizes, shapes, operating voltages, currents, and heat pumping capacities. The current trend leans towards employing a larger number of couples operating at lower currents. Before selecting an efficient device, certain parameters need to be evaluated:

TC: Temperature at the Cold Surface.

TH: Temperature at the Hot Surface.

TH encompasses two critical parameters:

1. The device's efficiency, which denotes the heat transfer between the hot surface of the TEC and the surrounding environment.
2. The temperature of the ambient environment into which the heat is dissipated. QC represents the heat absorbed at the Cold Surface. The object to be cooled is closely coupled with the cold surface of the TEC, causing its temperature to decrease until it reaches equilibrium with the TEC's cold surface temperature.

Now, ΔT can be defined as:

$$\Delta T = T_H - T_C$$

It's crucial to accurately determine this temperature difference (ΔT) to ensure the design operates as intended.

The ratio of QC to P, where QC represents the heat absorbed at the cold junction and P denotes the input power, is known as the Coefficient of Performance (COP) [5]. COP is commonly used to identify superior devices.

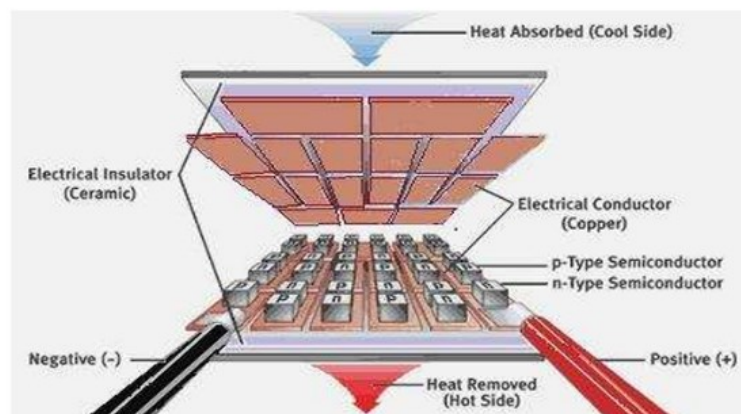


Figure 1: Overview of peltier device

Advantages of this device over conventional device:

Thermoelectric Coolers (TECs) offer numerous advantages over conventional refrigeration systems:

- They boast no moving parts, resulting in reduced maintenance requirements.
- TECs eliminate the need for chlorofluorocarbons, contributing to environmental preservation.
- Precise temperature control within fractions of degrees is achievable.
- Their flexible shape, or form factor, allows for compact designs, making them ideal for modern technological applications.

- TECs can function effectively in environments that are smaller or more demanding than those suitable for conventional refrigeration.
- With a mean time between failures (MTBF) exceeding almost 100,000 hours, TECs offer a long operational lifespan.
- They are easily controllable by adjusting input voltage or current.
- TECs draw comparatively lower current than compressor-based refrigeration systems, enhancing energy efficiency.

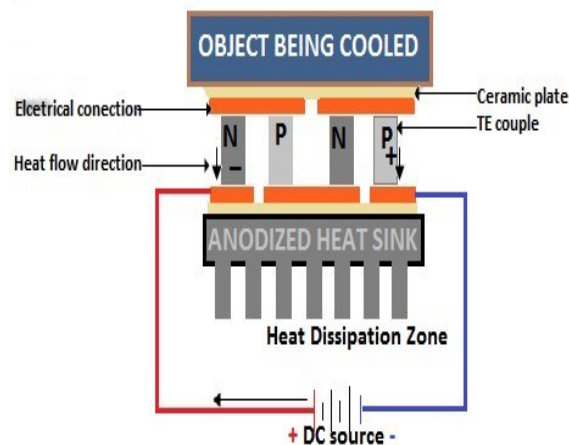


Figure 2: Working of Peltier Device

System assembly:

Achieving optimal performance with this device necessitates placing it in a relatively humid environment and directing airflow towards the cooler side of the TEC. This airflow allows water vapour to release its latent heat, reaching the dew point temperature and facilitating water condensation. To enhance this process, it's beneficial to circulate the same warm air through the cold side of the TEC to prevent water droplets from freezing and promote water generation. It's crucial to consider the purpose of the assembly, as the chosen technique is just as critical as selecting the appropriate device. All mechanical interfaces within the setup, including those of the objects being cooled and the ambient environment, must be addressed. Since mechanical interfaces inherently impede heat flow by adding thermal resistance, it's essential to optimize assembly techniques to minimize thermal resistance. When employing multiple modules between common plates, it's imperative to ensure that the variation in height between modules does not exceed 0.001" (please specify tolerance for lapped modules when placing orders). Typically, most Thermoelectric (TE) assemblies utilize thermal grease interfaces. The thickness of the grease should be constrained within 0.003 ± 0.0006 " [7]. To accommodate these tolerances effectively, specific cleanliness standards must be upheld. Grease has a high affinity for pollutants such as dirt, grit, and toxic powders like sulfur and carbon compounds. The presence of these contaminants can elevate impurity levels within the water flowing adjacent to the grease joints.

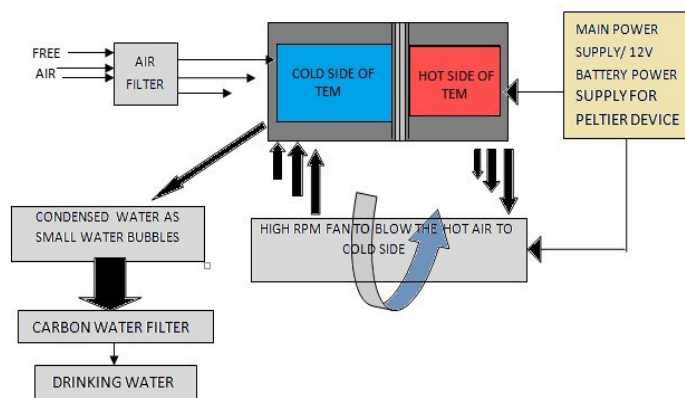


Figure 3: Schematic Diagram of System Assembly for Drinking Water

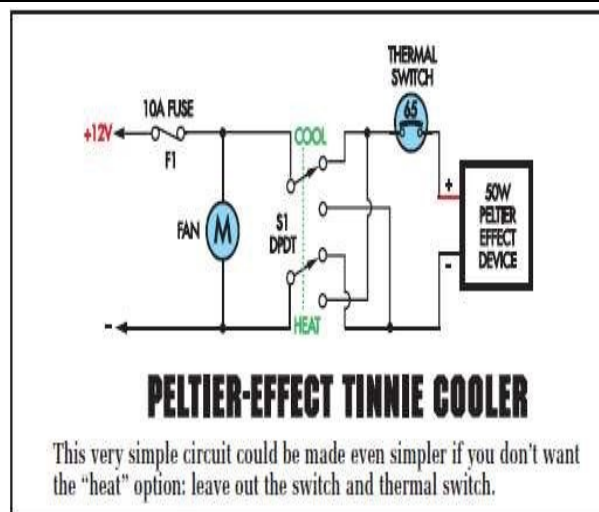


Figure 4: Simplest theoretical design of TEC assembly

Solar-driven thermoelectric:

Solar energy offers a significant advantage with its minimal water consumption during usage [8], making it particularly suitable for installation in regions facing freshwater scarcity. However, the adoption of solar refrigeration might have remained limited if not for the energy crisis of the 1970s, which spurred modern reconsideration [9]. During this period, research into Peltier's cooling effect and the Photovoltaic effect gained traction, primarily driven by initiatives like the World Health Organization's cold chain project and international health organizations' efforts in rural areas [10]. Solar cells were integrated into small thermoelectrically operated refrigerators for these initiatives. Experimental investigations and analyses on solar cell-driven thermoelectric refrigerators were conducted [11]. The key components of the solar battery include the Photo Voltaic (PV) array, the storage battery, and the controller. The PV array, installed outdoors, harnesses solar energy, while the battery stores excess energy generated during periods of sunlight. This stored energy is then utilized to power the system during nighttime hours. The controller, typically equipped with a microprocessor, monitors battery status and power consumption, adjusting system operation accordingly. Its architecture or circuit design is relatively straightforward, primarily focused on safeguarding the battery against overcharging or overdischarging. Specially designed lead-acid batteries, suitable for deep discharge cycles, are often employed in such systems, although other battery types may also be suitable.

Required calculations and equations: One of the primary goals from the outset of this project has been to attain the dew point temperature and exercise precise control over temperature to efficiently condense water. Achieving accurate temperature control on the cold side is paramount. The equation for calculating the dew point temperature is given as follows [2]: $T_{dp} = (Hr/100)^{(1/8)} * (112 + 0.9T_a) + 0.1T_a - 112$ (2) Here, T_{dp} ($^{\circ}C$) represents the dew point temperature, Hr (%) denotes relative humidity, and T_a ($^{\circ}C$) signifies the ambient air temperature to be condensed, i.e., the air within the cold chamber (as illustrated in the diagram). TEC1 modules are typically arranged side by side and enclosed between two ceramic plates. The cooling capacity of this configuration is directly proportional to the number of TEC1 modules present. The quantity of heat absorbed correlates with the current and duration: $W = P * I * t$ (3) In this equation, P represents the Peltier Coefficient, I denotes the current, P_{serve} as the foundation for advancing this prototype project.

Working of the device:

Utilizing the gathered data about the ambient environment—including Hr and T_a —the dew point temperature (T_{dp}) is calculated. Subsequently, the Peltier coefficient (P) and current rating of the TEC1 device can be determined from the device's data table. This information facilitates the estimation of the time required to generate water droplets from the humid air. Upon activation, the hot side of the device begins to heat up while the cold side cools, eventually reaching the dew point temperature. The cold side of the TEC1 module cools the air passing through its heat sink area, causing water vapours to condense, akin to the condensation process observed outside a glass filled with ice. It's noteworthy that while the TEC initiates quickly to reach the dew

point temperature, it may take some time initially to start producing water. However, with time, the process accelerates to yield sufficient water, as evidenced by the experimental results of this project. TEC1 modules are selected for their moderate efficiency requirements and their ability to achieve optimum cooling effects over time. Since this project primarily relies on solar energy, employing higher-rated TECs may not be immediately feasible due to the higher voltage requirements. Thus, opting for TEC1 modules, despite their slightly longer operational duration, proves to be advantageous.

III. EQUIPMENTS AND RESULT

The solar cell unit integrated into this system is rated for 12 V output with a maximum power output of 120 W, providing sufficient power to sustain three 40 W (3.5 A) Peltier coolers (TEC1) connected in parallel. Each Peltier device measures 4x4x0.8 cm and is capable of generating a maximum temperature difference (ΔT) of 87°C. To facilitate air circulation, a 3000rpm fan measuring 15x15 cm (matching the size of the TEC1) is employed, capable of producing airflow of up to 500 cubic feet per minute (cfm). The heat sinks utilized are crafted from anodized aluminium for optimal heat dissipation, while activated carbon filters are employed for water filtration purposes. Temperature control is achieved through the integration of two temperature sensors and either a PIC16F872 or ATmega series microcontroller.

Explanation of AWG:

The amount of water vapour air can hold depends on two things: temperature and humidity. Warmer air can hold more water vapour than cold air. In this device, the moist air is first passed through a tube that's cooled by a Peltier element. This cooling process helps the air hold less water vapour.

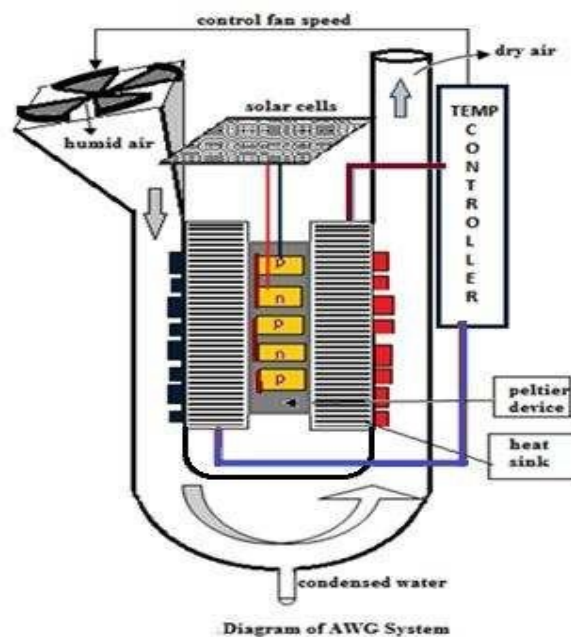


Figure 5: Project design of Atmospheric Water

Generator

As the cool air leaves the tube, it can't hold as much water vapour anymore, so the moisture condenses into water droplets. The air then travels to the hot side of the Peltier element, which is important to keep the device from overheating. This warm air can actually be recycled back to the cool side to improve water production, as mentioned earlier. The collected water is then stored in a container and may be optionally filtered. Testing this device involves monitoring temperature and humidity. By measuring these factors at the air inlet and outlet, along with air flow rate, we can create a control system. This system, often built with specific sensors and microcontrollers, keeps the circulating air above freezing to prevent ice formation. It also controls fan speed and overall system safety based on temperature readings from both sides of the Peltier element. While exact output can vary, estimates suggest this system can produce around 1 liter of condensed water per hour under ideal daytime conditions.

IV. APPLICATIONS

It's nearly impossible to disregard the portability of this device, given its straightforward design and durability. Hence, the Atmospheric Water Generator emerges as an indispensable tool for addressing extreme situations, including flood relief efforts, desert regions, and rural areas. Offering considerable advantages, it functions as a renewable source of atmospheric water and operates without the need for a robust power source. Several companies, such as 'Water-maker India Ltd' and 'Aerowater,' have already introduced similar devices for domestic use. Moreover, its application extends to industrial development, where water scarcity poses a significant challenge.

V. CONCLUSION AND FUTURE WORK

When deployed in highly humid regions, this system demonstrates the capability to produce approximately 1 liter of condensed water per hour during daylight hours, showcasing promising potential. This success paves the way for the design of an enhanced system featuring advanced solar cells with greater power output and the ability to store excess energy generated during the day for night-time use. However, the economic viability of such systems remains somewhat uncertain due to their relatively high installation costs.

Nevertheless, there are avenues for future expansion: 1) Large-scale implementation could incorporate Reverse Osmosis (RO) and Ultraviolet (UV) water filters to ensure the produced water meets the standards set by organizations like the WHO and BIS. 2) Exploration of more efficient models of Peltier devices beyond TEC1 could enhance system performance.

3) While the current project prioritizes portability, larger-sized scrubbers for improved air filtration could be considered for comprehensive implementation, capable of removing all airborne oxides.

4) Furthermore, the project's concept presents a promising alternative to conventional refrigeration systems, hinting at a transformative shift in cooling engineering. The utilization of low-power semiconductor devices suggests an impending evolution in cooling technology, challenging preconceived notions about the energy consumption associated with refrigeration science. Consequently, the future holds the potential for widespread adoption of such devices, currently confined to experimental settings.

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